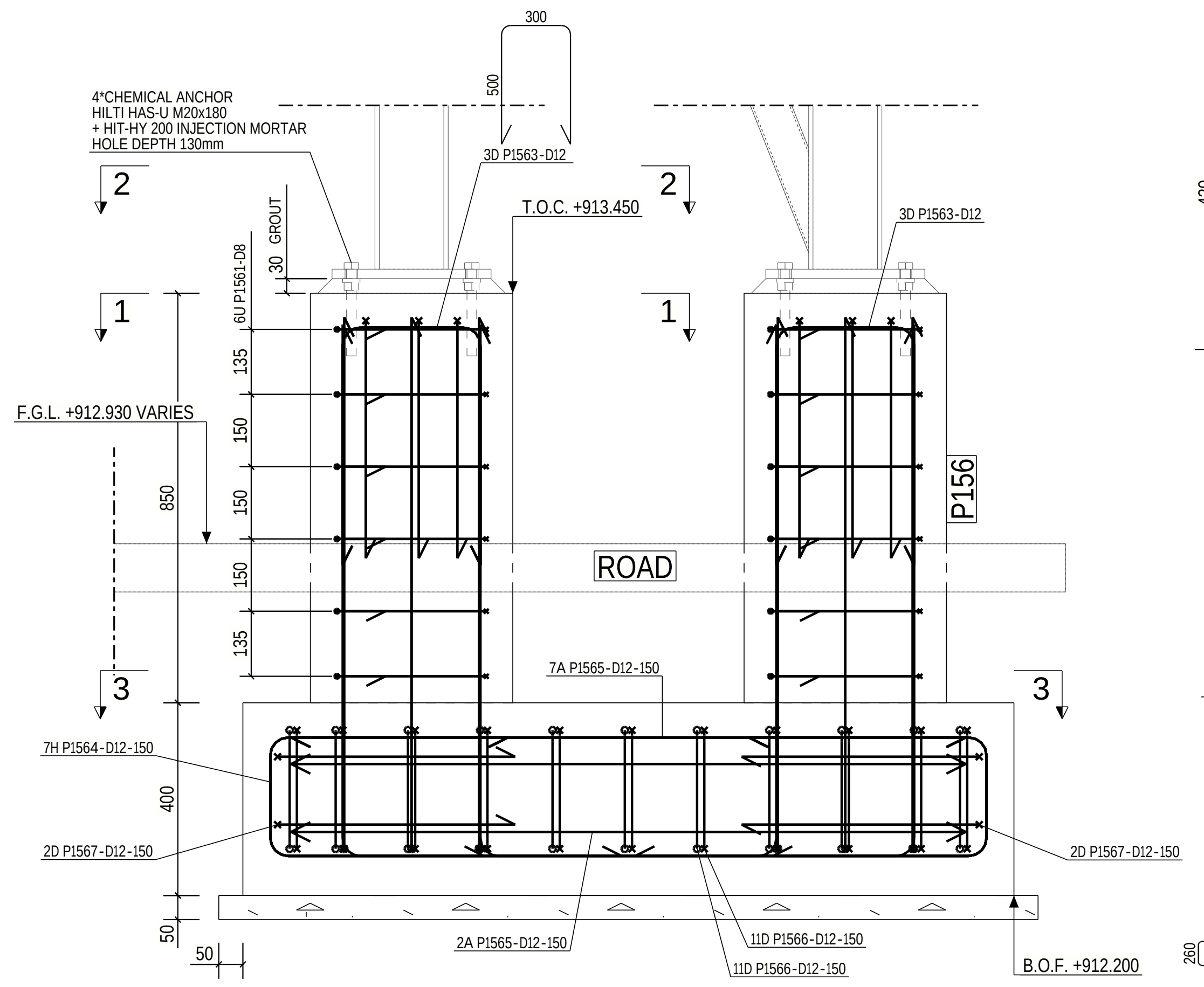
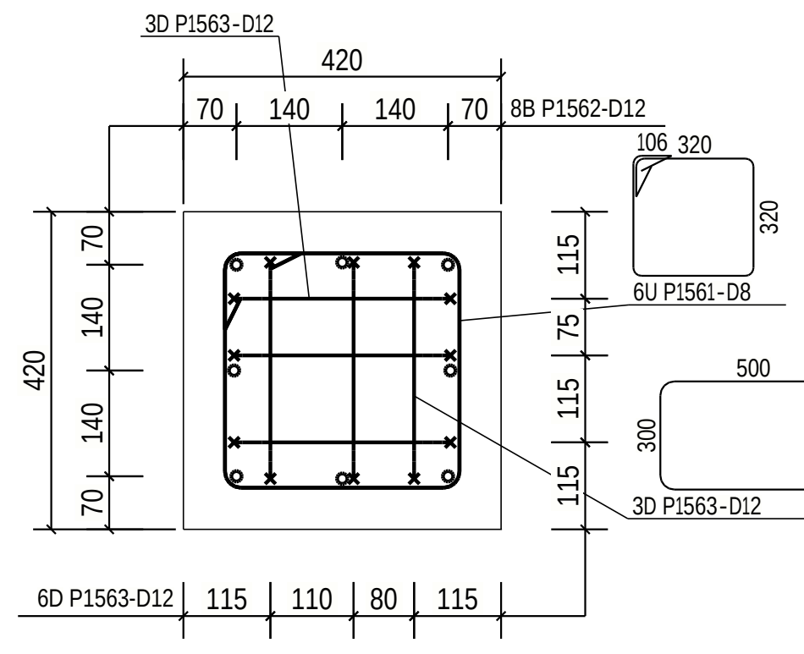
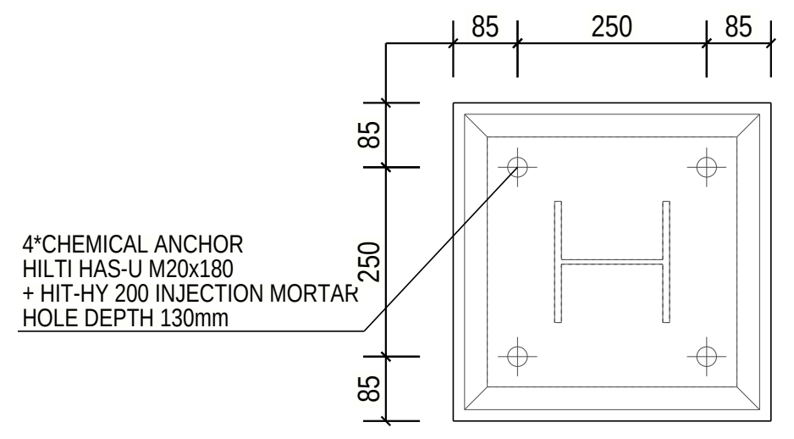
MD-216-49W1-EG-CV-DED-1037
DRAWING NUMBER
REV D01


4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
3D P1563-D12
T.O.C. +913.450
F.G.L. +912.930 VARIES
ROAD
P156
7A P1565-D12-150
7H P1564-D12-150
2D P1567-D12-150
2A P1565-D12-150
11D P1566-D12-150
B.O.F. +912.200
1
2
3

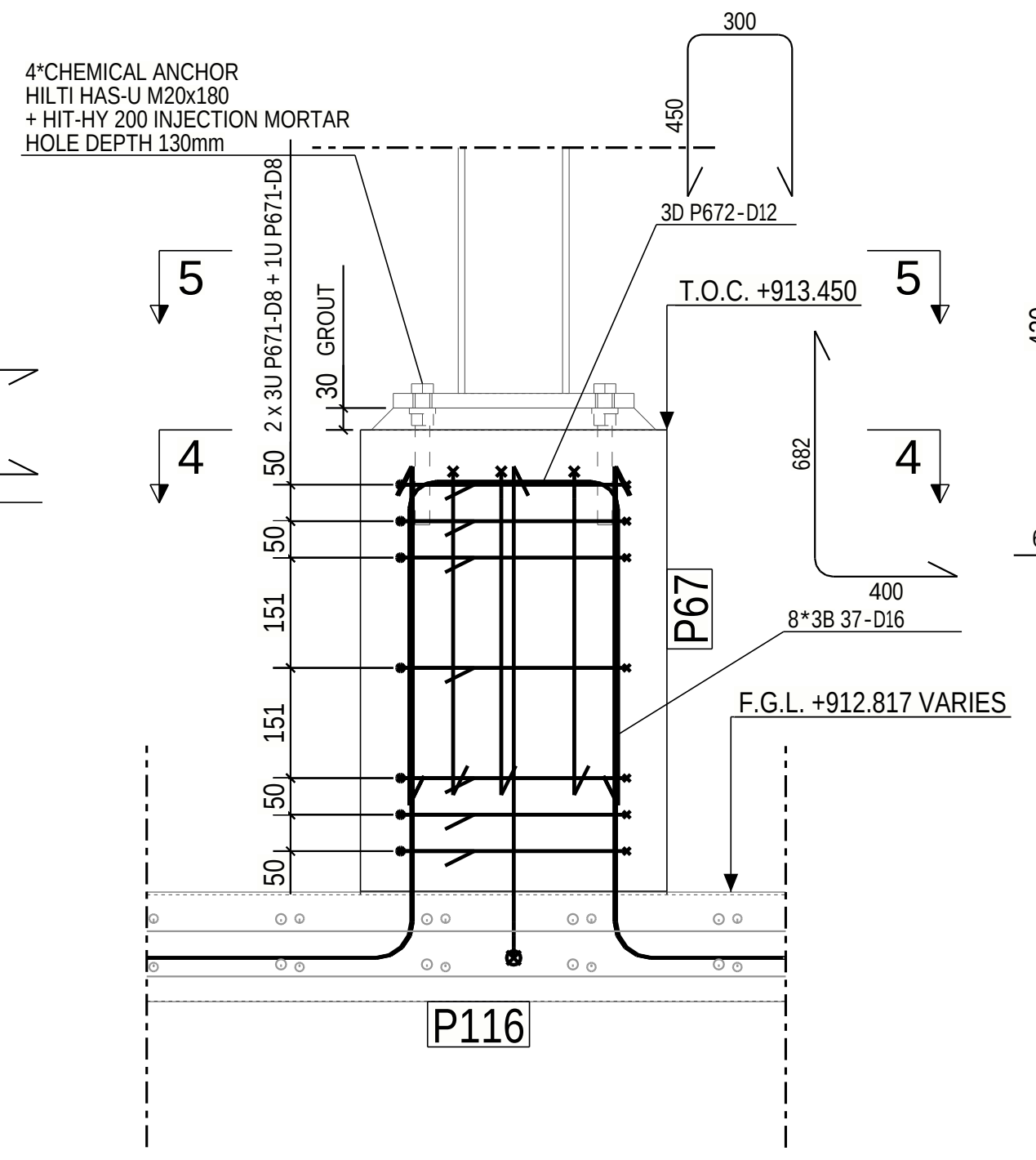
FOUNDATION P156 REINFORCEMENT
(SCALE 1:10)


3D P1563-D12
420
70 140 140 70
8B P1562-D12
6U P1561-D8
3D P1563-D12
6D P1563-D12 115 110 80 115

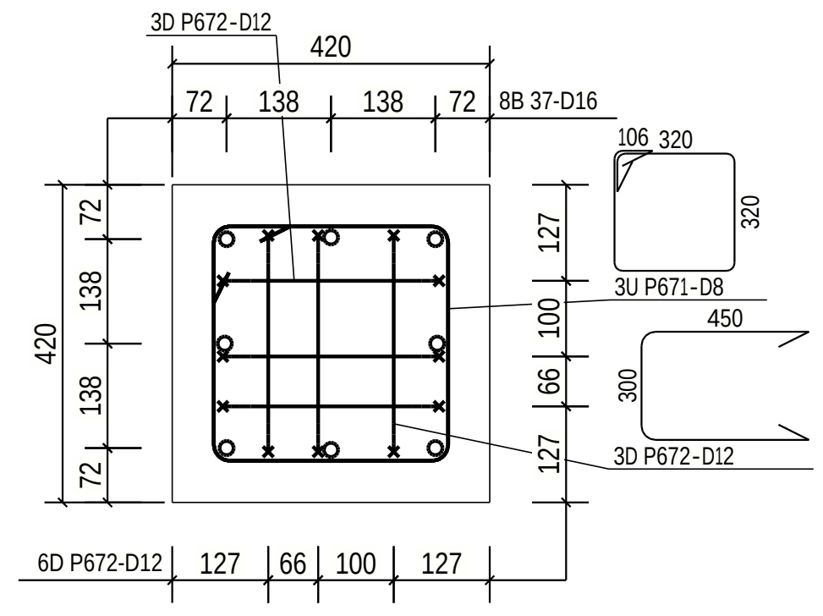
SECTION : 1 - 1
(SCALE 1:10)


85 250 85
4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm

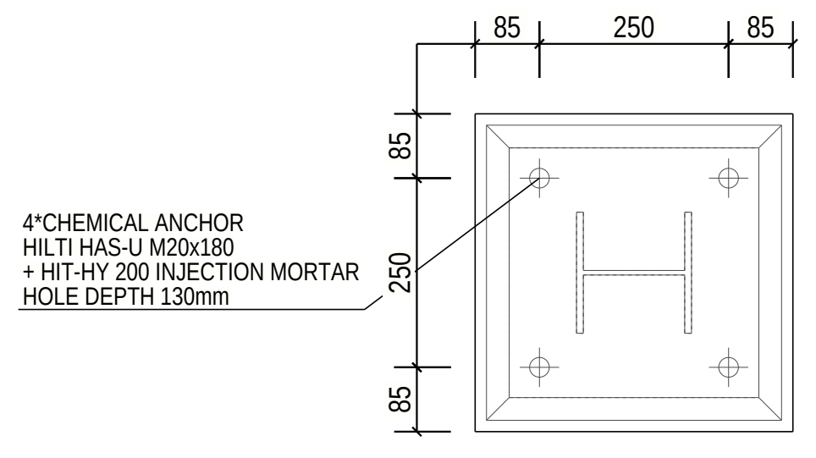
SECTION : 2 - 2
(SCALE 1:10)


4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
3D P672-D12
T.O.C. +913.450
8*38 37-D16
F.G.L. +912.817 VARIES
P67
P116
4
5

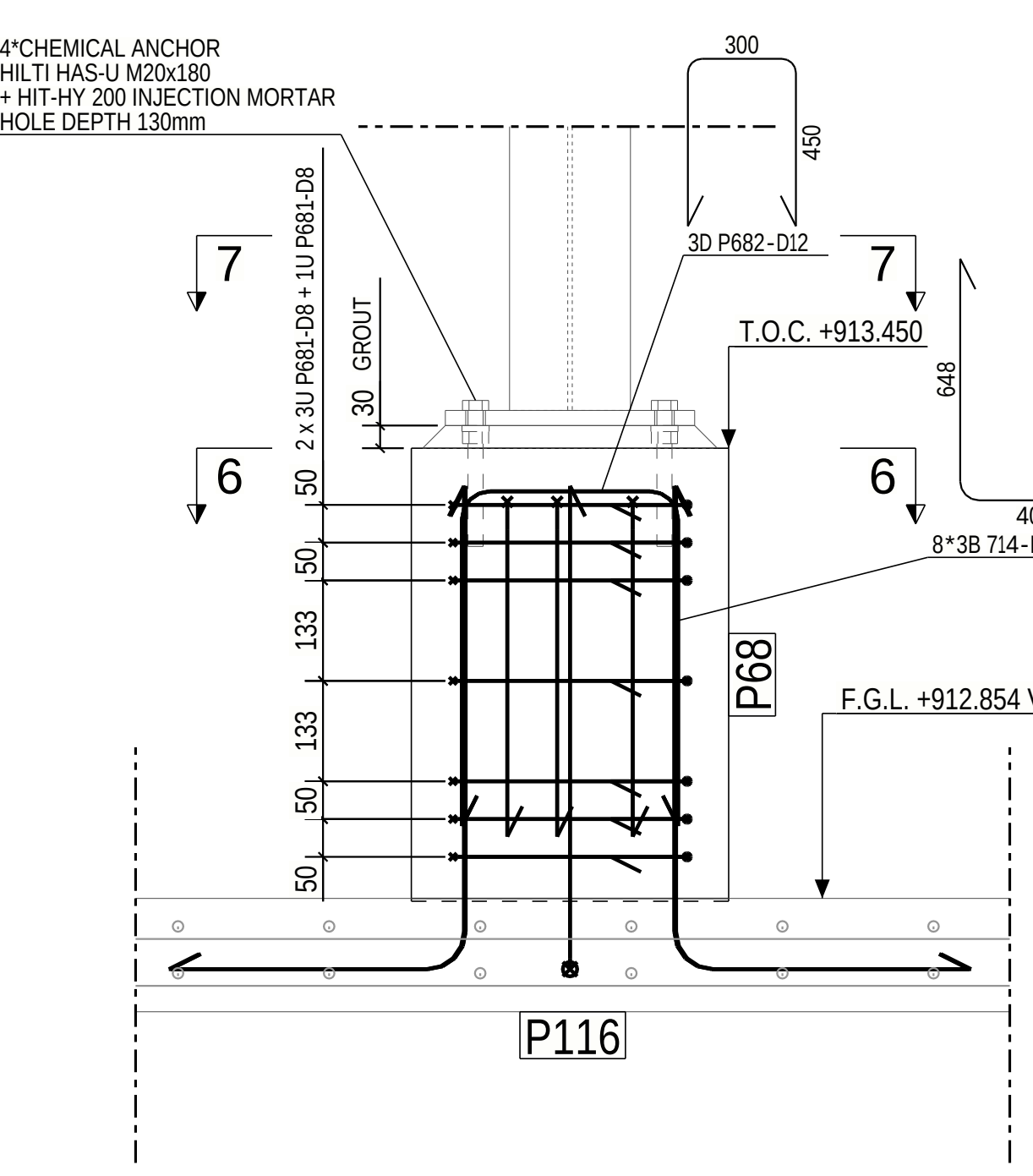
FOUNDATION P67 REINFORCEMENT
(SCALE 1:10)


3D P672-D12
72 138 138 72
8B 37-D16
3U P671-D8
3D P672-D12
6D P672-D12 127 66 100 127

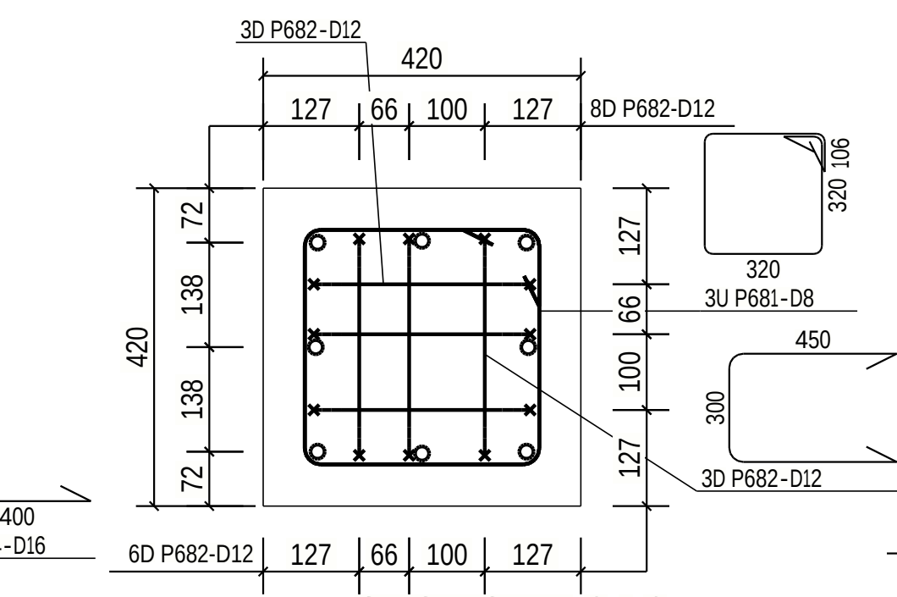
SECTION : 4 - 4
(SCALE 1:10)


85 250 85
4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm

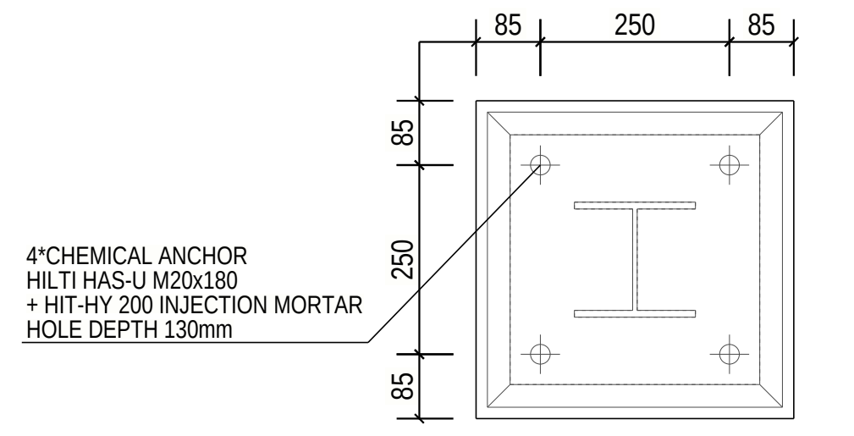
SECTION : 5 - 5
(SCALE 1:10)


4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
3D P682-D12
T.O.C. +913.450
8*38 7I4-D16
F.G.L. +912.854 VARIES
P68
P116
6
7

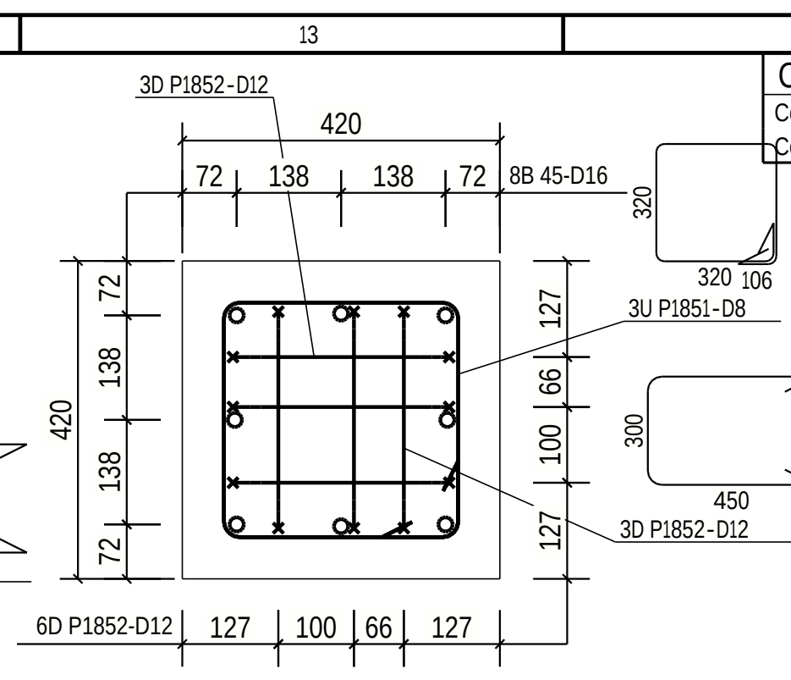
FOUNDATION P68 REINFORCEMENT
(SCALE 1:10)


3D P682-D12
127 66 100 127
8D P682-D12
3U P681-D8
3D P682-D12
6D P682-D12

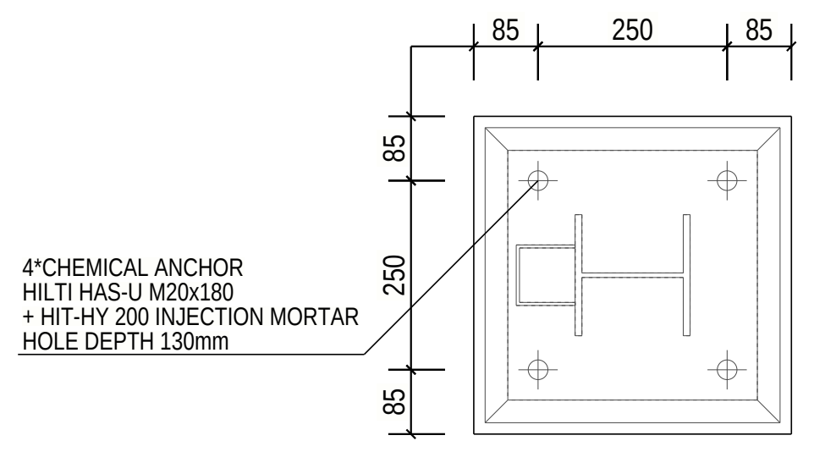
SECTION : 6 - 6
(SCALE 1:10)


85 250 85
4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm

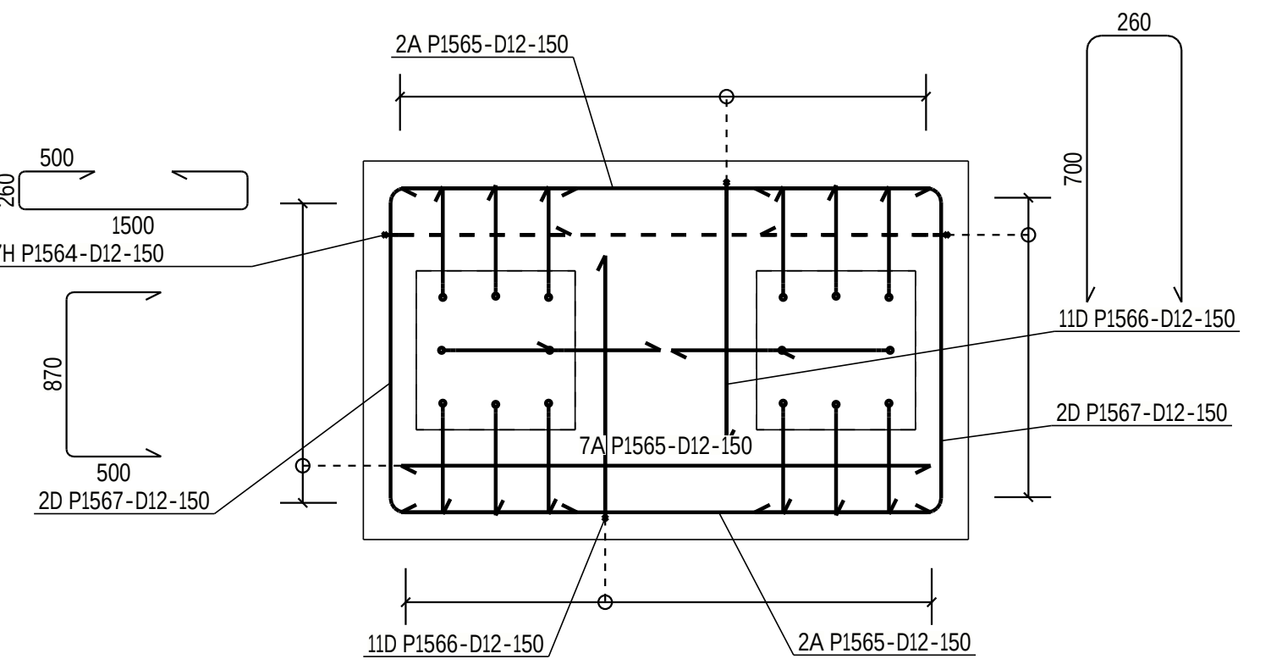
SECTION : 7 - 7
(SCALE 1:10)


3D P1852-D12
72 138 138 72
8B 45-D16
3U P1851-D8
3D P1852-D12
6D P1852-D12 127 100 66 127

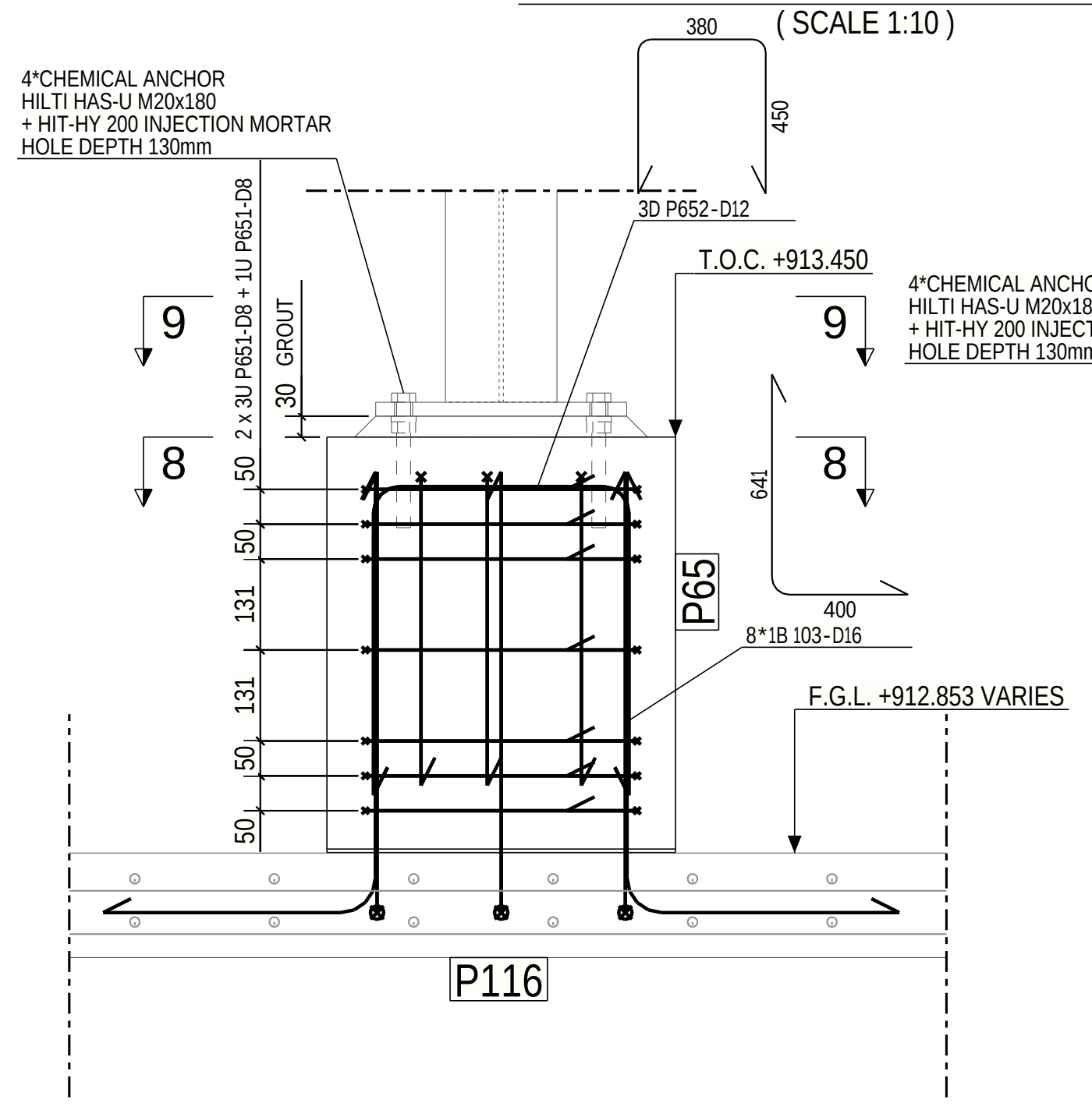
SECTION : 11 - 11
(SCALE 1:10)


85 250 85
4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm

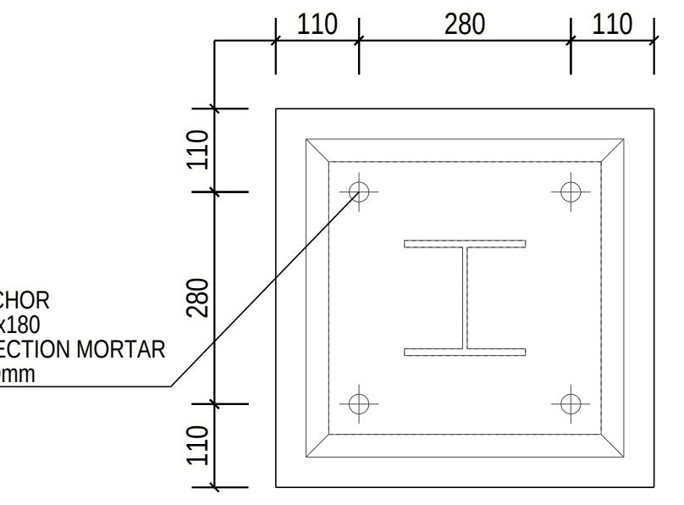
SECTION : 12 - 12
(SCALE 1:10)


2A P1565-D12-150
1500
7H P1564-D12-150
11D P1566-D12-150
2D P1567-D12-150
7A P1565-D12-150
11D P1566-D12-150
2A P1565-D12-150

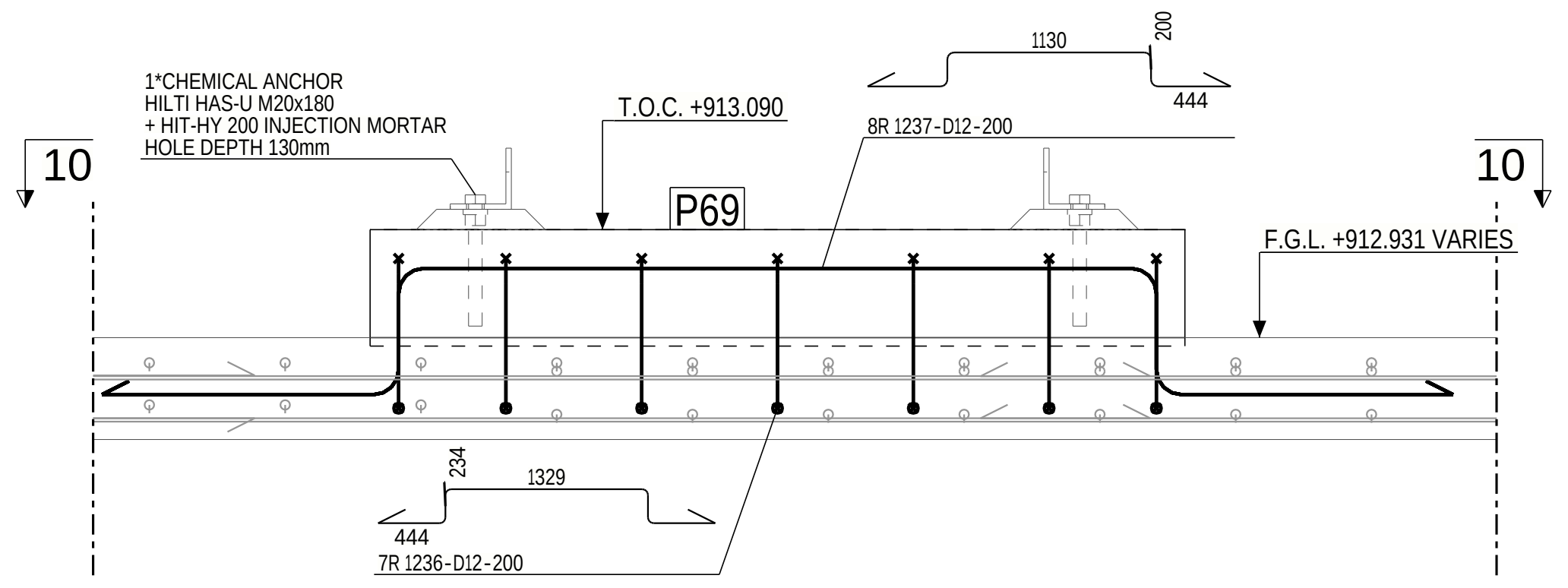
SECTION : 3 - 3
(SCALE 1:20)


4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
3D P652-D12
T.O.C. +913.450
8*18 103-D16
F.G.L. +912.853 VARIES
P65
P116
8
9

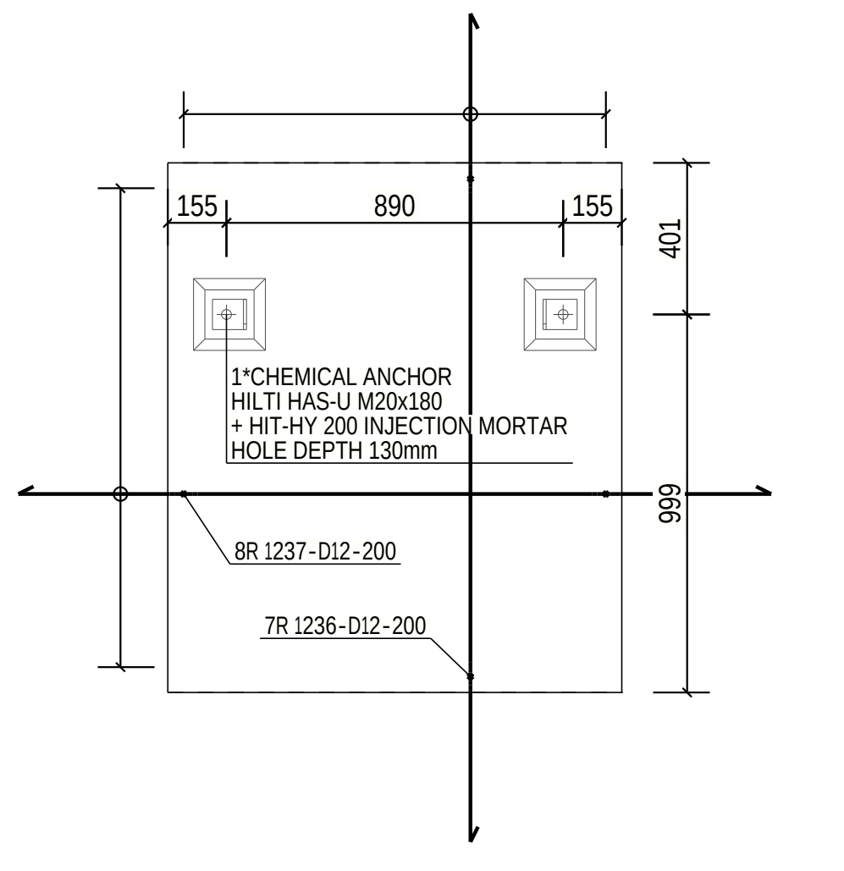
FOUNDATION P65 REINFORCEMENT
(SCALE 1:10)


110 280 110
4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm

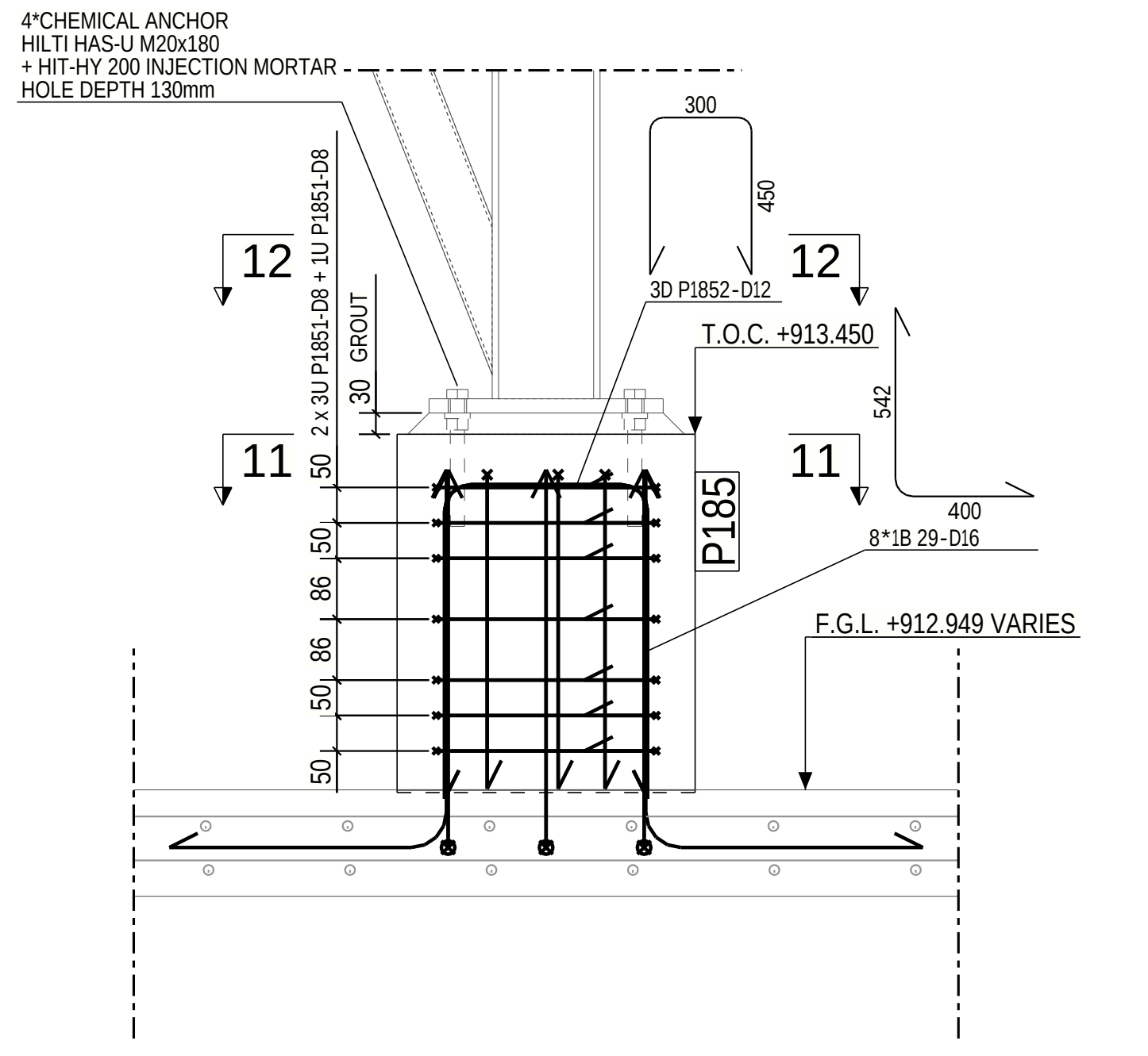
SECTION : 9 - 9
(SCALE 1:10)


1*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
T.O.C. +913.090
8R 1237-D12-200
F.G.L. +912.931 VARIES
P69
7R 1236-D12-200
10

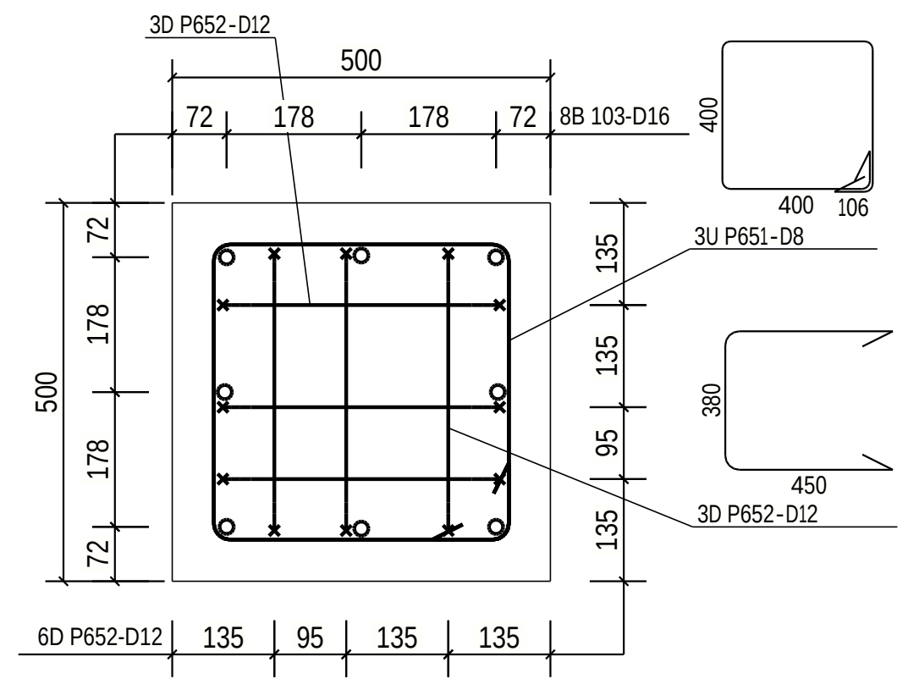
STAIR LANDING P69 REINFORCEMENT
(SCALE 1:10)


155 890 155
1*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
8R 1237-D12-200
7R 1236-D12-200

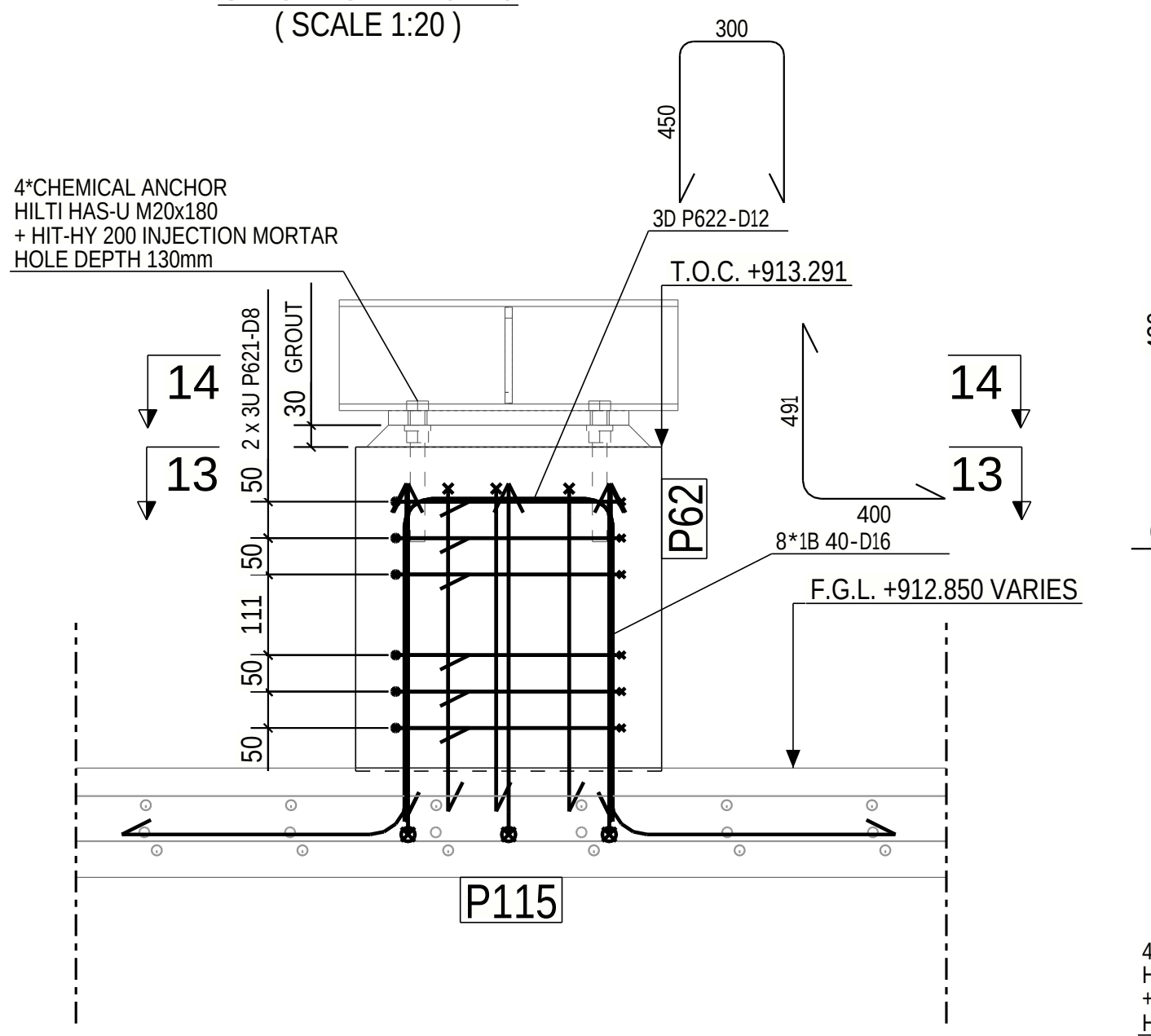
SECTION : 10 - 10
(SCALE 1:20)


4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
3D P1852-D12
T.O.C. +913.450
8*18 29-D16
F.G.L. +912.949 VARIES
P185
11
12

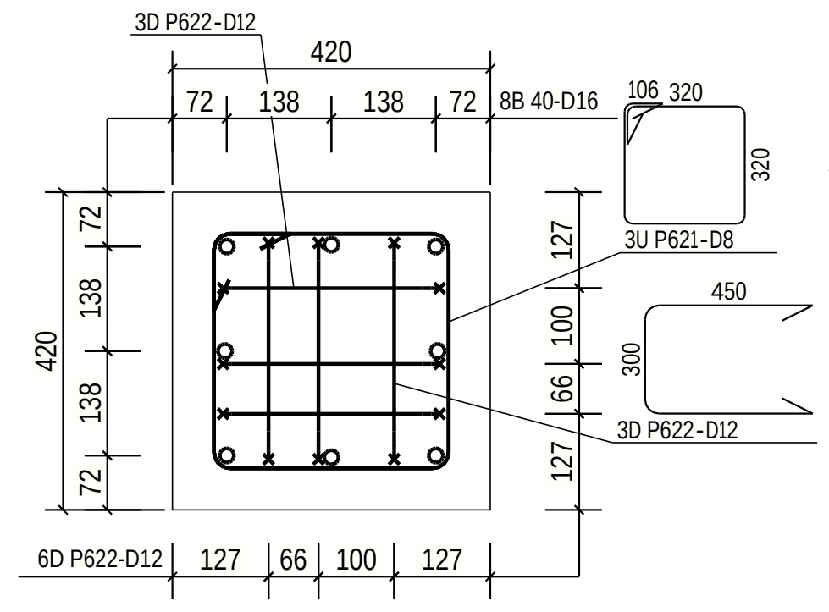
FOUNDATION P185 REINFORCEMENT
(SCALE 1:10)


3D P652-D12
500
72 178 178 72
8B 103-D16
3U P651-D8
3D P652-D12
6D P652-D12 135 95 135 135

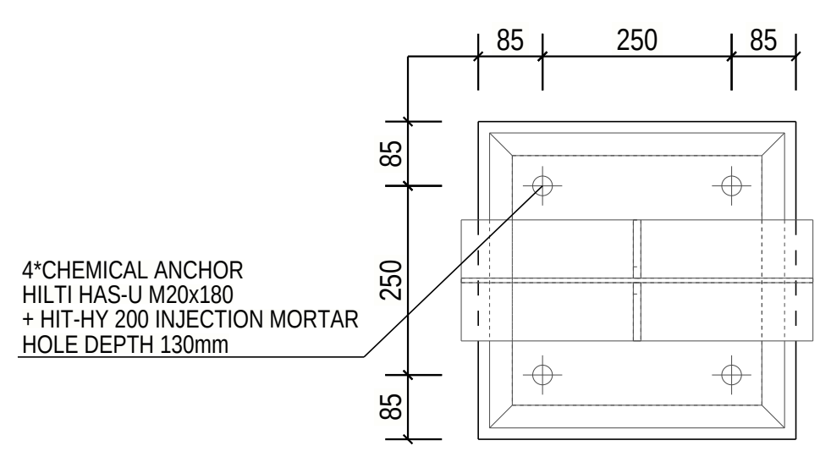
SECTION : 8 - 8
(SCALE 1:10)


4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
3D P622-D12
T.O.C. +913.291
8*18 40-D16
F.G.L. +912.850 VARIES
P62
P115
13
14

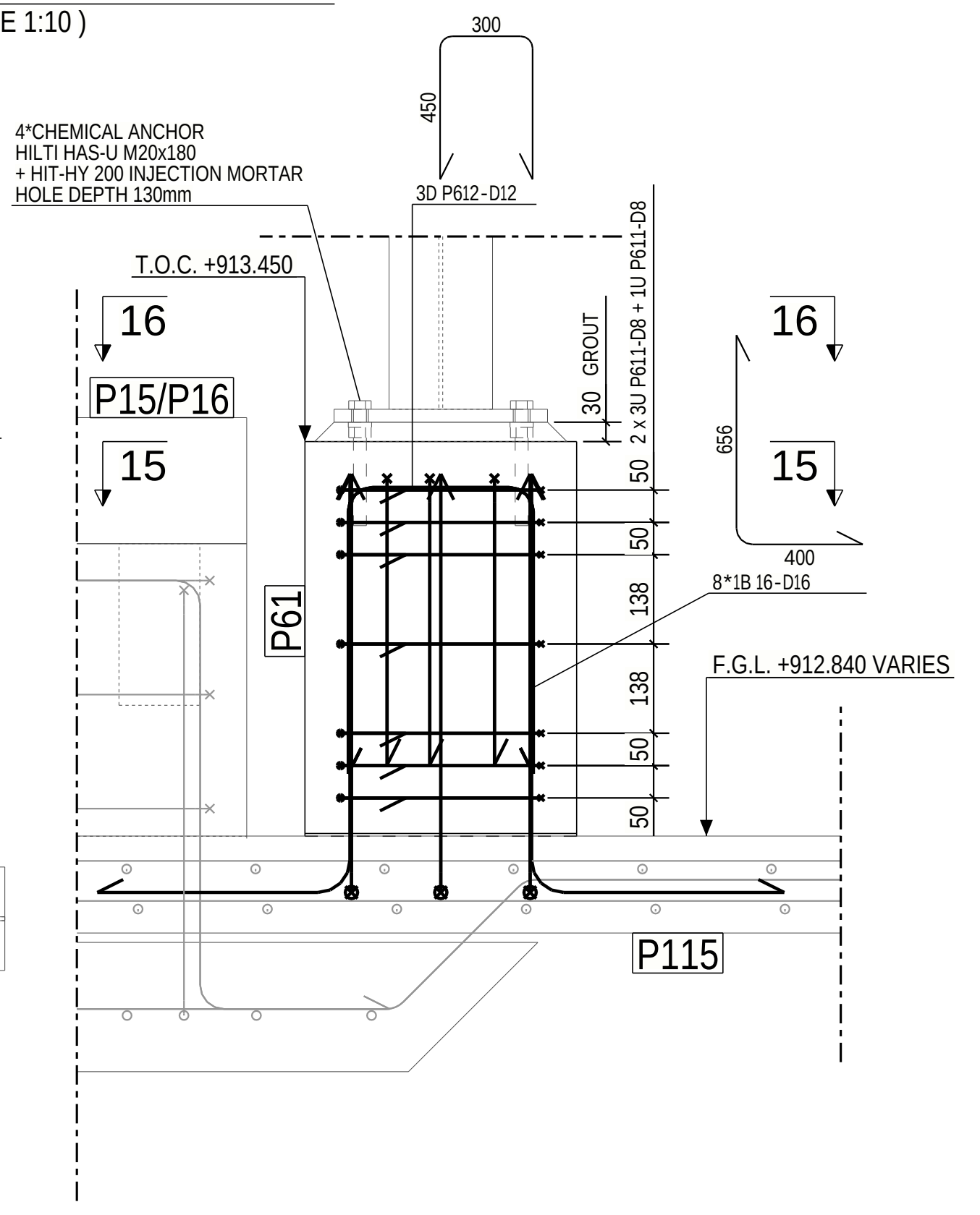
FOUNDATION P62 REINFORCEMENT
(SCALE 1:10)


3D P622-D12
420
72 138 138 72
8B 40-D16
3U P621-D8
3D P622-D12
6D P622-D12 127 66 100 127

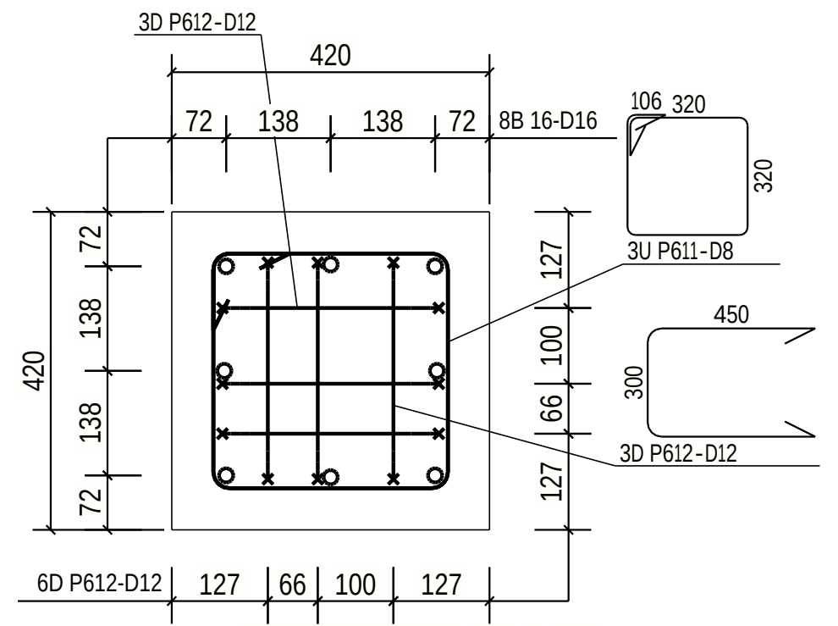
SECTION : 13 - 13
(SCALE 1:10)


85 250 85
4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm

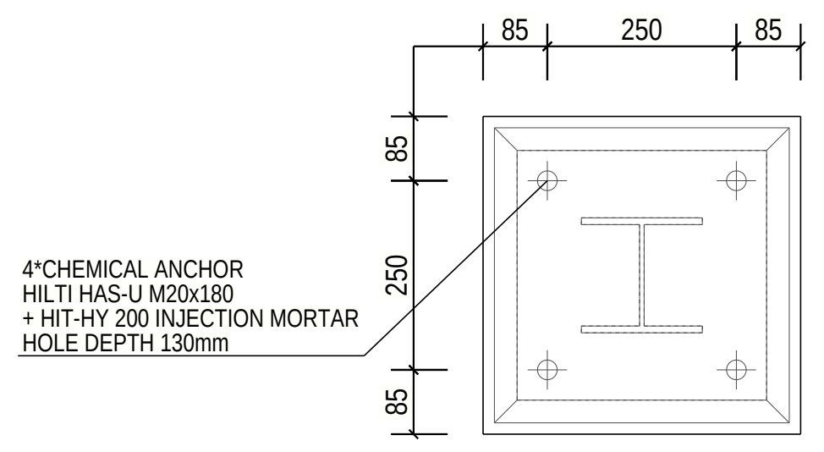
SECTION : 14 - 14
(SCALE 1:10)


4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
T.O.C. +913.450
3D P612-D12
8*18 16-D16
F.G.L. +912.840 VARIES
P15/P16
P61
P115
15
16

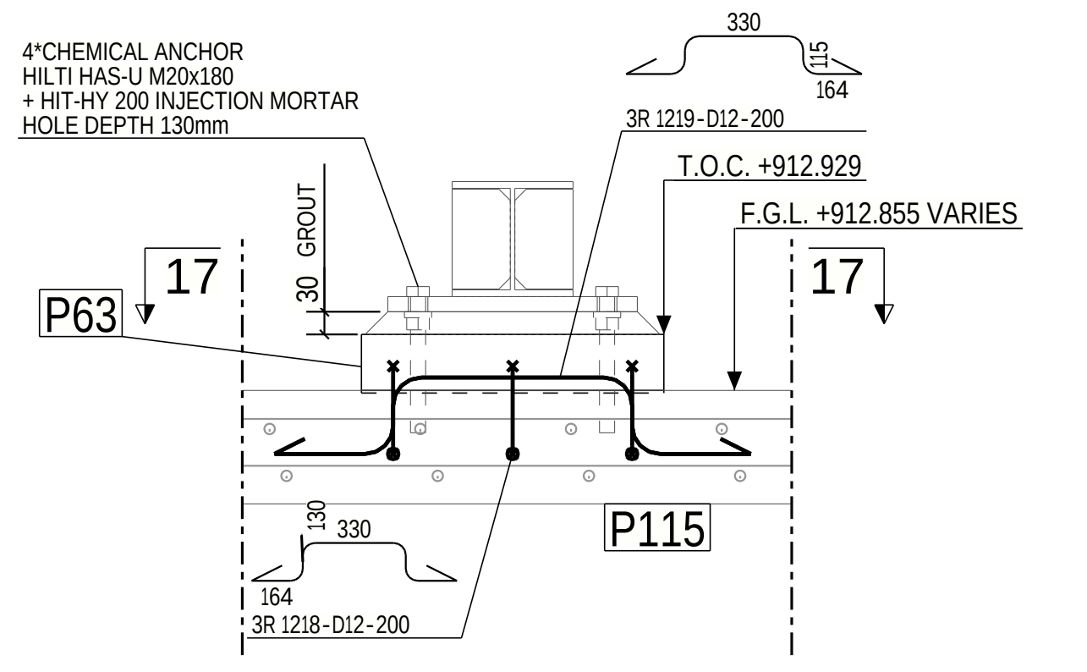
FOUNDATION P61 REINFORCEMENT
(SCALE 1:10)


3D P612-D12
420
72 138 138 72
8B 16-D16
3U P611-D8
3D P612-D12
6D P612-D12 127 66 100 127

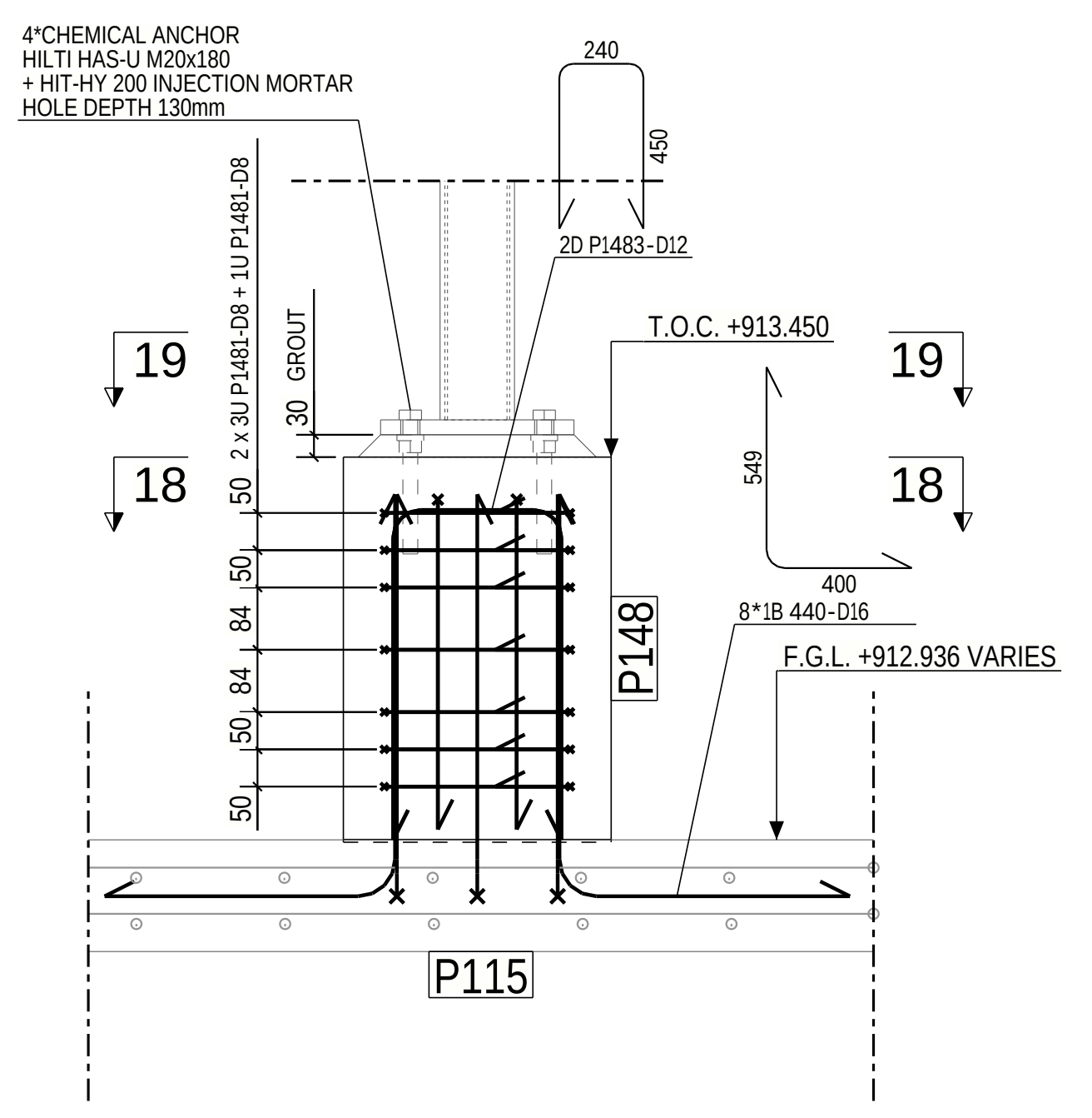
SECTION : 15 - 15
(SCALE 1:10)


85 250 85
4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm

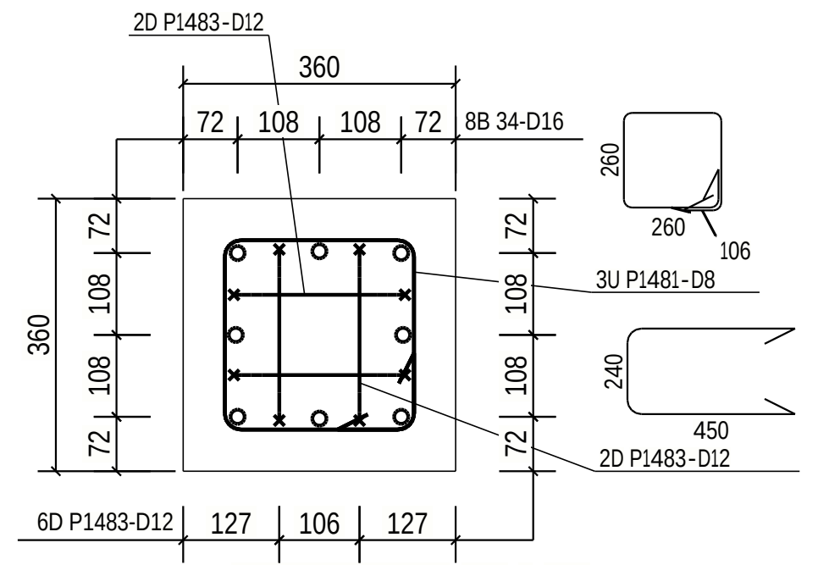
SECTION : 16 - 16
(SCALE 1:10)


4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
3R 1219-D12-200
T.O.C. +912.929
F.G.L. +912.855 VARIES
P63
P115
3R 1218-D12-200
17

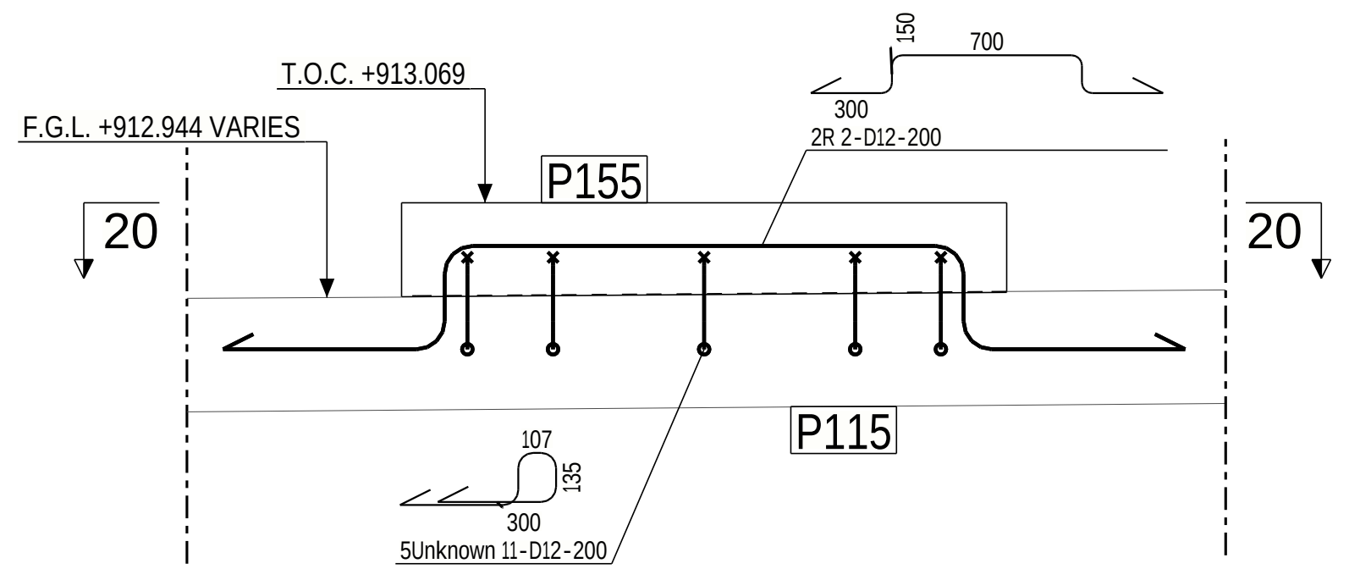
FOUNDATION P63 REINFORCEMENT
(SCALE 1:10)


4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
2D P1483-D12
T.O.C. +913.450
8*18 440-D16
F.G.L. +912.936 VARIES
P148
P115
18
19

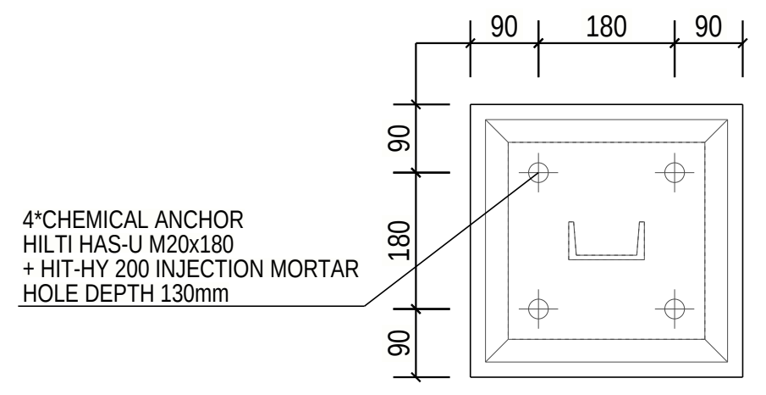
FOUNDATION P148 REINFORCEMENT
(SCALE 1:10)


2D P1483-D12
360
72 108 108 72
8B 34-D16
3U P1481-D8
2D P1483-D12
6D P1483-D12 127 106 127

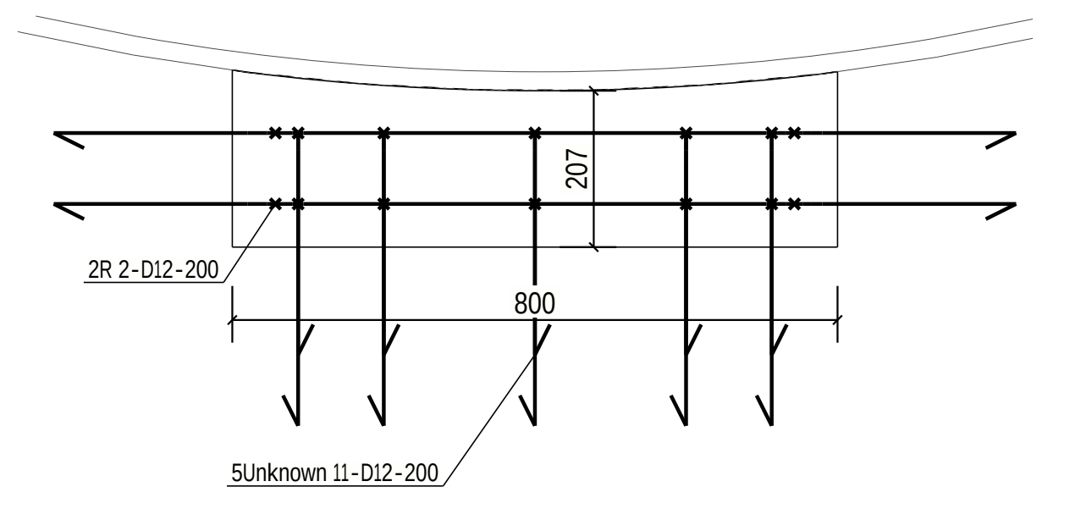
SECTION : 18 - 18
(SCALE 1:10)


T.O.C. +913.069
F.G.L. +912.944 VARIES
2R 2-D12-200
P155
P115
5Unknown 11-D12-200
20

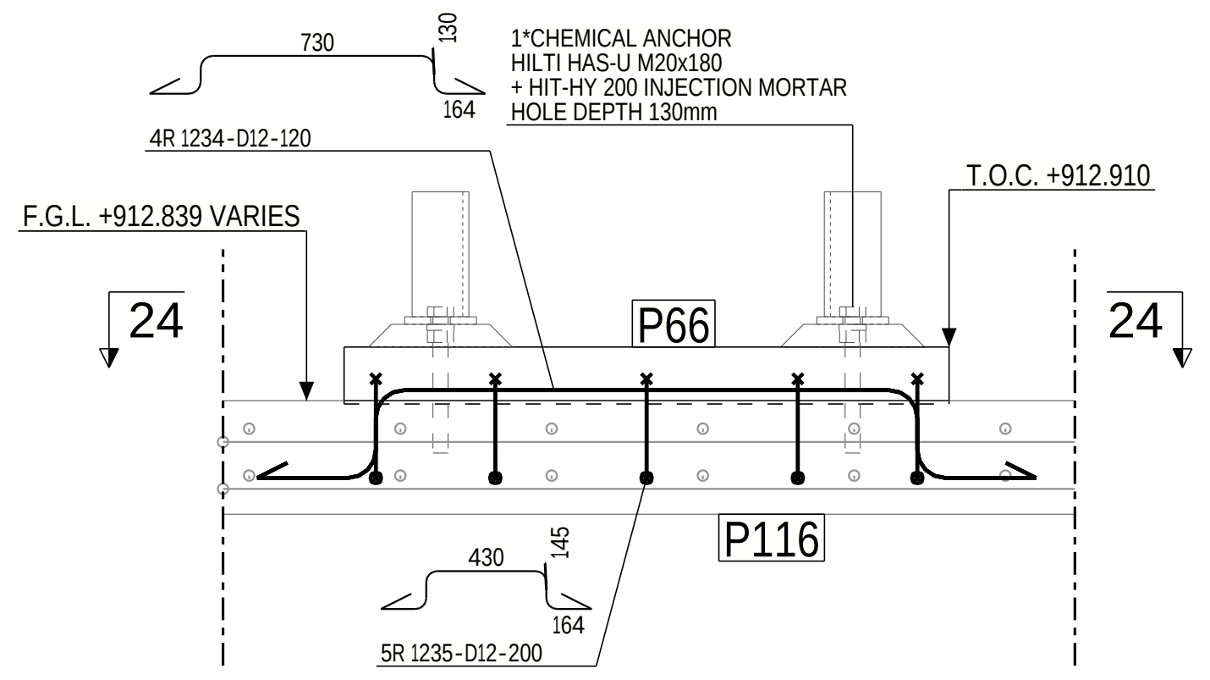
LADDER PLATFORM P155 REINFORCEMENT
(SCALE 1:10)


90 180 90
4*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm

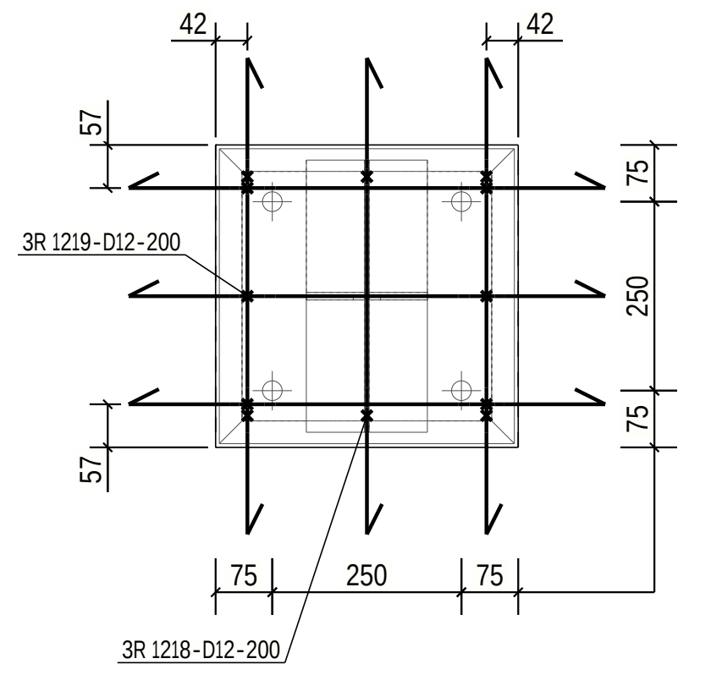
SECTION : 19 - 19
(SCALE 1:10)


2R 2-D12-200
800
5Unknown 11-D12-200

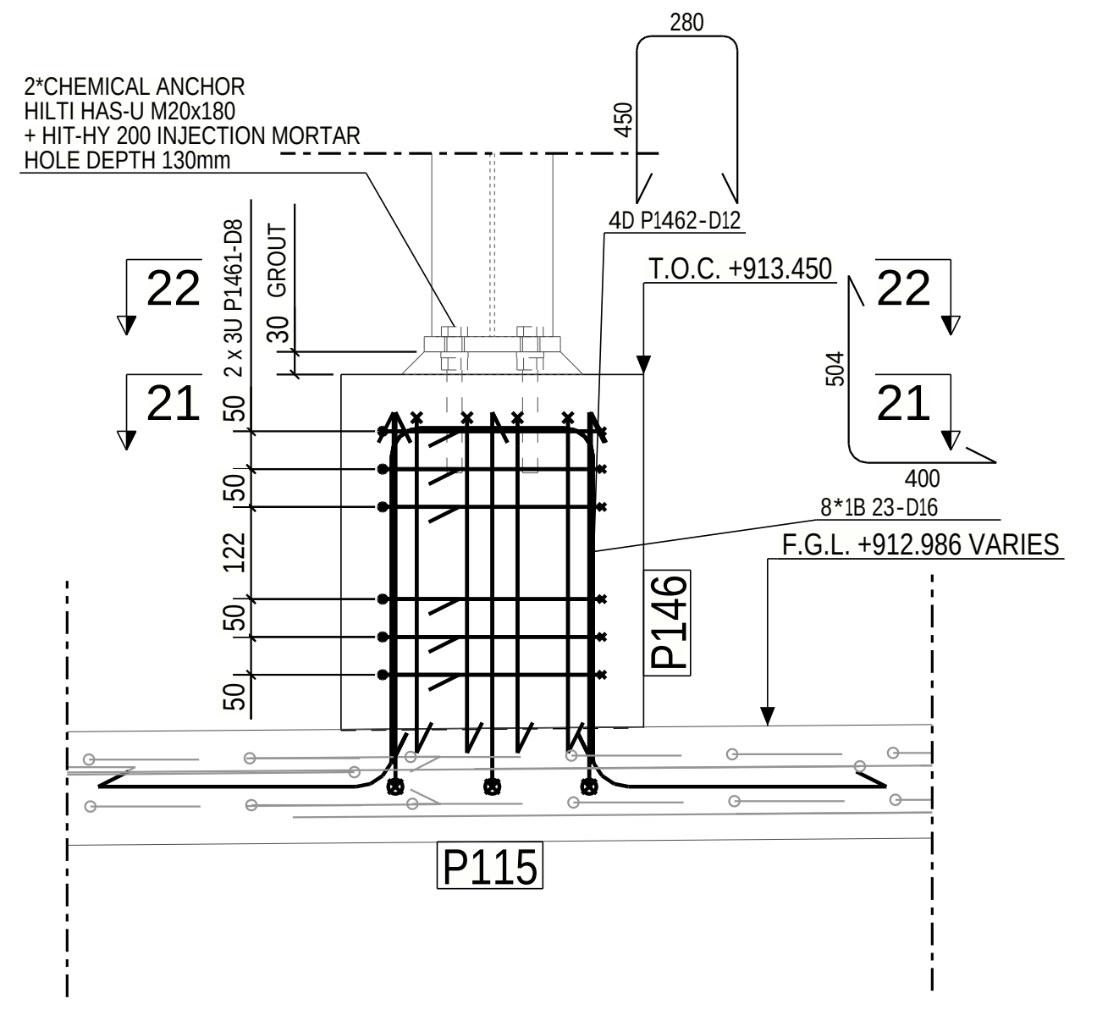
SECTION : 20 - 20
(SCALE 1:10)


4R 1234-D12-120
1*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
F.G.L. +912.839 VARIES
T.O.C. +912.910
P66
P116
5R 1235-D12-200
24

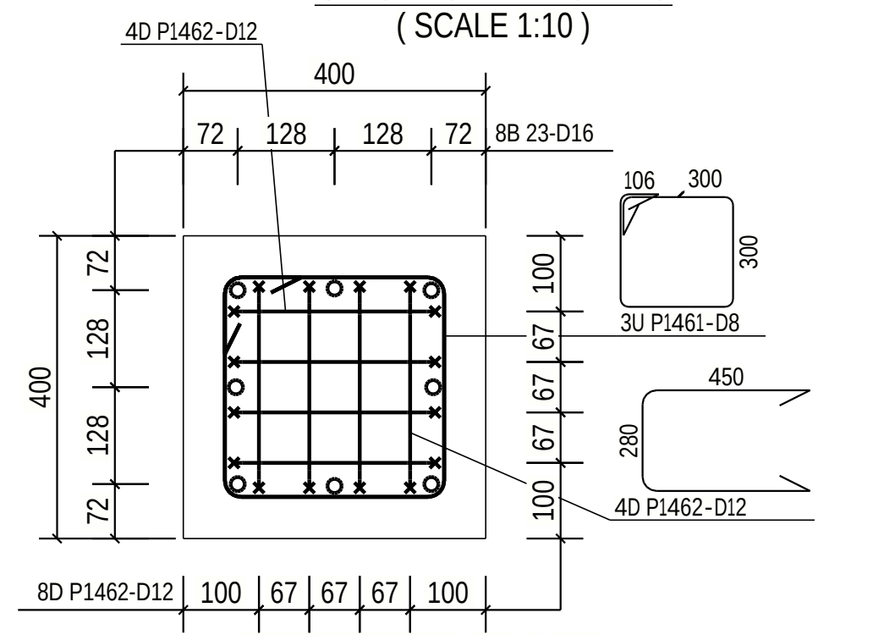
FOUNDATION P66 REINFORCEMENT
(SCALE 1:10)


3R 1219-D12-200
75 250 75
3R 1218-D12-200

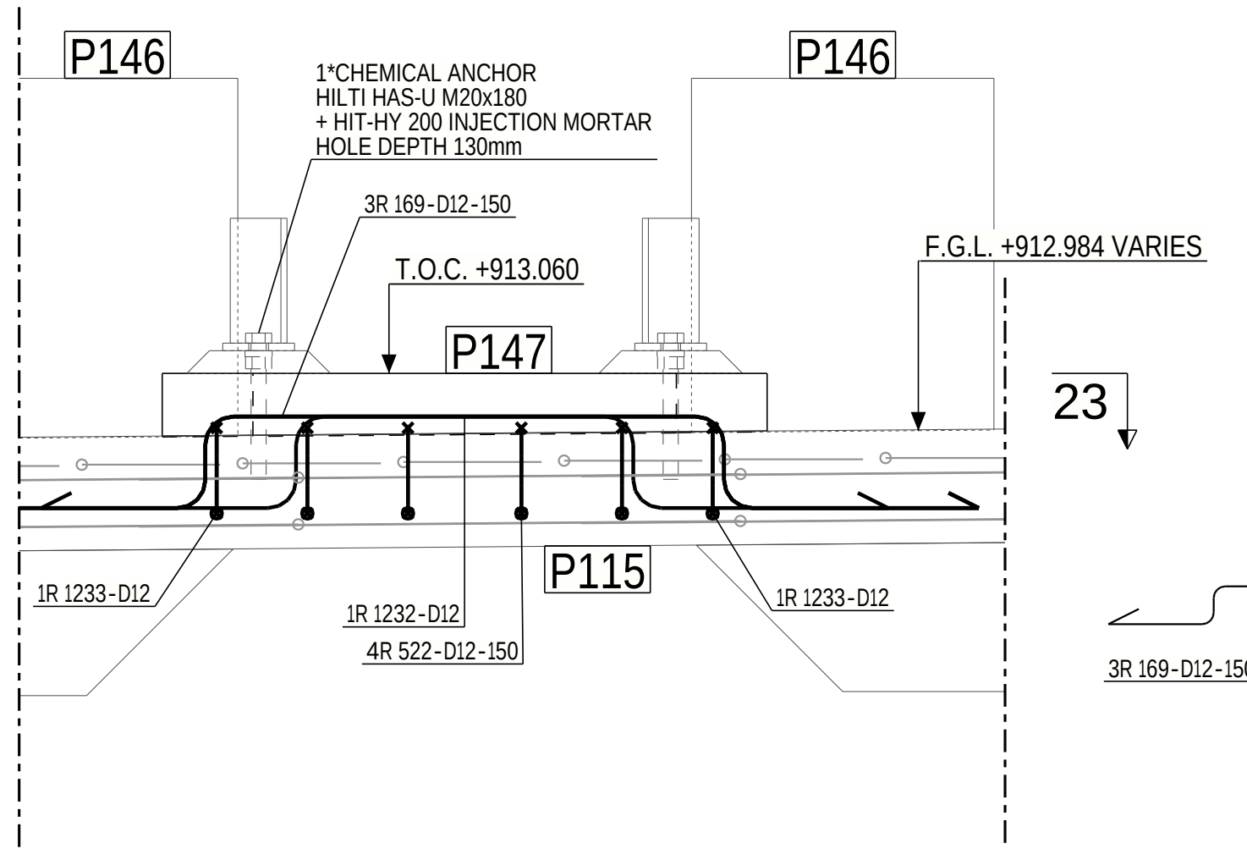
SECTION : 17 - 17
(SCALE 1:10)


2*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
4D P1462-D12
T.O.C. +913.450
8*18 23-D16
F.G.L. +912.986 VARIES
P146
P115
21
22

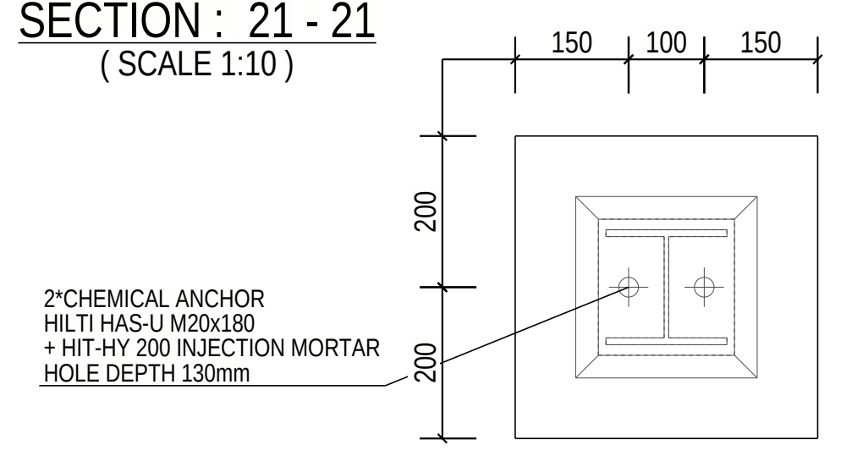
FOUNDATION P146 REINFORCEMENT
(SCALE 1:10)


4D P1462-D12
400
72 128 128 72
8B 23-D16
3U P1461-D8
4D P1462-D12
8D P1462-D12 100 67 67 67 100

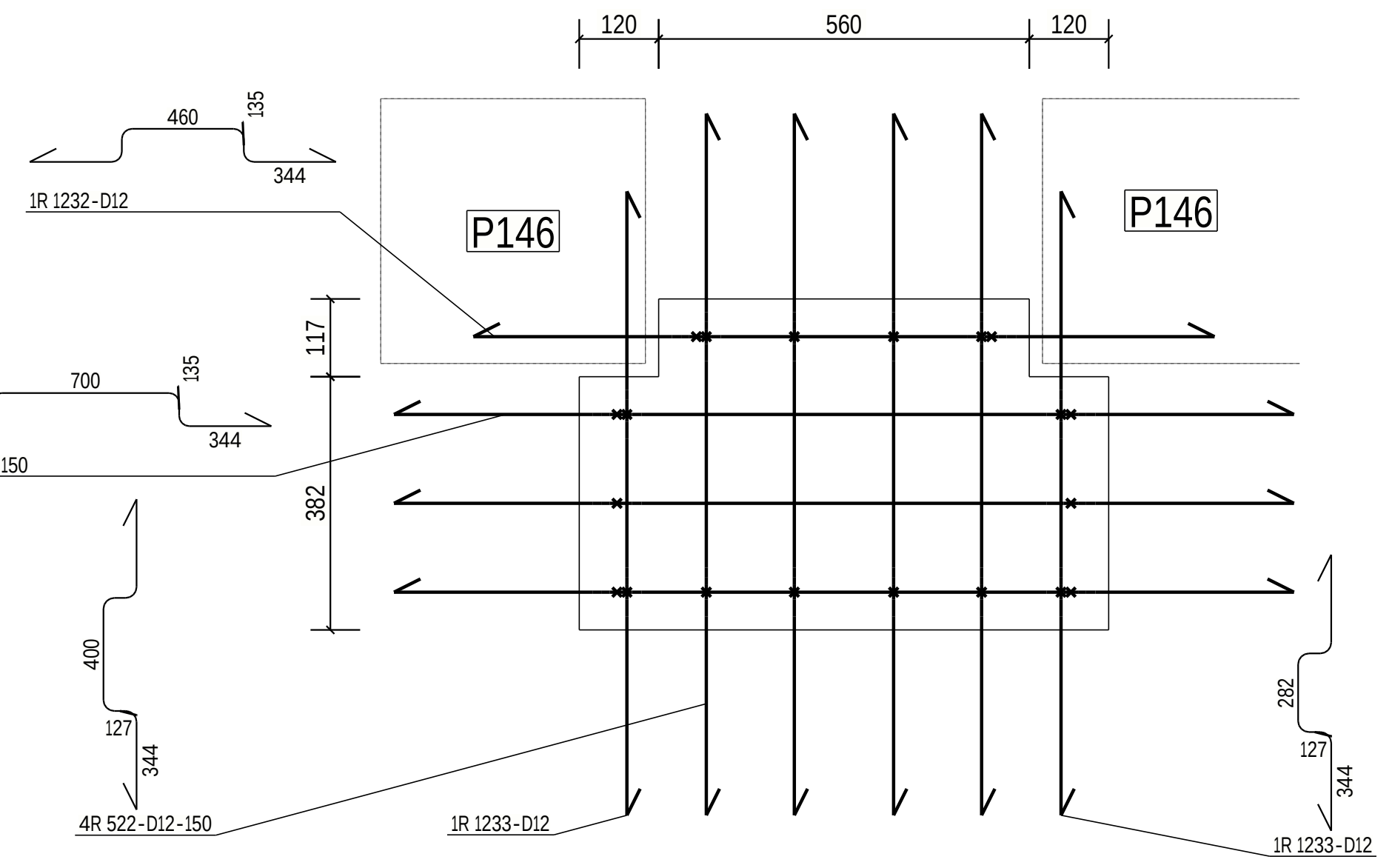
SECTION : 21 - 21
(SCALE 1:10)


P146
1*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
3R 169-D12-150
T.O.C. +913.060
F.G.L. +912.984 VARIES
P147
P115
1R 1233-D12
1R 1232-D12
4R 522-D12-150
23

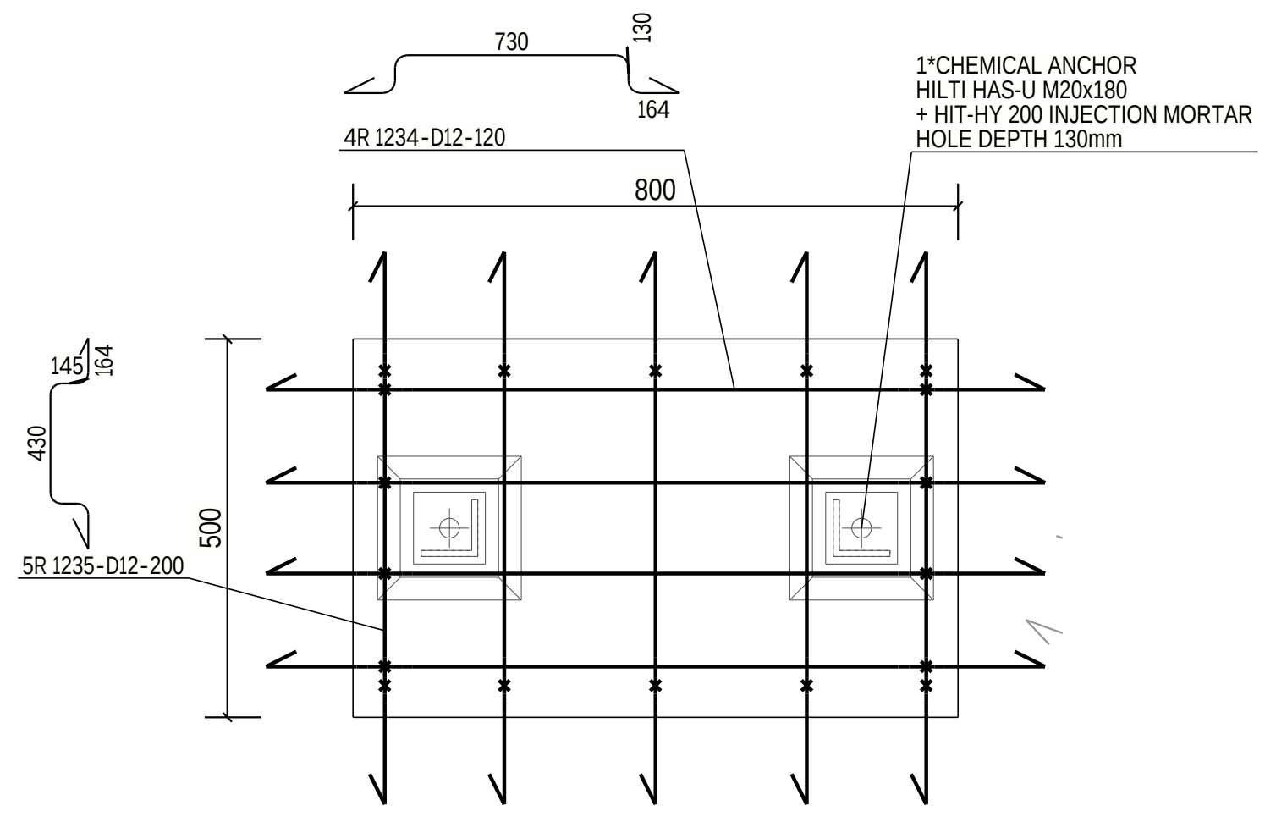
LADDER PLATFORM P147 REINFORCEMENT
(SCALE 1:10)


150 100 150
2*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm

SECTION : 22 - 22
(SCALE 1:10)


120 560 120
1R 1232-D12
P146
3R 169-D12-150
4R 522-D12-150
1R 1233-D12

SECTION : 23 - 23
(SCALE 1:10)


4R 1234-D12-120
1*CHEMICAL ANCHOR HILTI HAS-U M20x180 + HIT-HY 200 INJECTION MORTAR HOLE DEPTH 130mm
800
5R 1235-D12-200

SECTION : 24 - 24
(SCALE 1:10)

NOTE: LOCATIONS FOR STARTER BARS ARE SHOWN IN DRAWINGS:
MD-216-49W1-EG-CV-DED-1027
MD-216-49W1-EG-CV-DED-1026
AND NOT SHOWN IN REINFORCEMENT LIST

LOCATIONS FOR STRUCTURES IN GENERAL ARRANGEMENT DRAWINGS:
MD-216-49W1-EG-CV-GAD-1001
MD-216-49W1-EG-CV-GAD-1002

CONCRETE INFORMATION

Concretes used	C35
Concrete cover 1	50±10mm

REINFORCING BAR LIST — CAST UNIT CODE: P61 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P611	7	B500B	8	1390		3.9	106	320	320	320	320				32	
D	P612	6	B500B	12	1140		6.1	450	300	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.9

REINFORCING BAR LIST — CAST UNIT CODE: P62 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P621	6	B500B	8	1390		3.3	106	320	320	320	320				32	
D	P622	6	B500B	12	1140		6.1	450	300	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.4

REINFORCING BAR LIST — CAST UNIT CODE: P63 — PCS: 1

REINFORCING BAR LIST — CAST UNIT CODE: P65 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P651	7	B500B	8	1710		4.7	106	400	400	400	400				32	
D	P652	6	B500B	12	1220		6.5	450	380	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 11.2

REINFORCING BAR LIST — CAST UNIT CODE: P66 — PCS: 1

REINFORCING BAR LIST — CAST UNIT CODE: P67 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P671	7	B500B	8	1390		3.9	106	320	320	320	320				32	
D	P672	6	B500B	12	1140		6.1	450	300	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.9

REINFORCING BAR LIST — CAST UNIT CODE: P68 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P681	7	B500B	8	1390		3.9	106	320	320	320	320				32	
D	P682	6	B500B	12	1140		6.1	450	300	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.9

REINFORCING BAR LIST — CAST UNIT CODE: P69 — PCS: 1

REINFORCING BAR LIST — CAST UNIT CODE: P146 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1461	6	B500B	8	1310		3.1	106	300	300	300	300				32	
D	P1462	8	B500B	12	1120		8.0	450	280	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 11.1

REINFORCING BAR LIST — CAST UNIT CODE: P147 — PCS: 1

REINFORCING BAR LIST — CAST UNIT CODE: P148 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1481	7	B500B	8	1150		3.2	106	260	260	260	260				32	
D	P1483	4	B500B	12	1080		3.8	450	240	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 7.0

REINFORCING BAR LIST — CAST UNIT CODE: P155 — PCS: 1

REINFORCING BAR LIST — CAST UNIT CODE: P156 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1561	12	B500B	8	1390		6.6	106	320	320	320	320				32	
B	P1562	16	B500B	12	1380		19.6	300	1110				90			60	
D	P1563	12	B500B	12	1240		13.2	500	300	500						60	
H	P1564	7	B500B	12	2900		18.0	500	260	1500	260	500				60	
A	P1565	11	B500B	12	1400		13.7	1400									
D	P1566	22	B500B	12	1600		31.3	700	260	700						60	
D	P1567	4	B500B	12	1810		6.4	500	870	500						60	

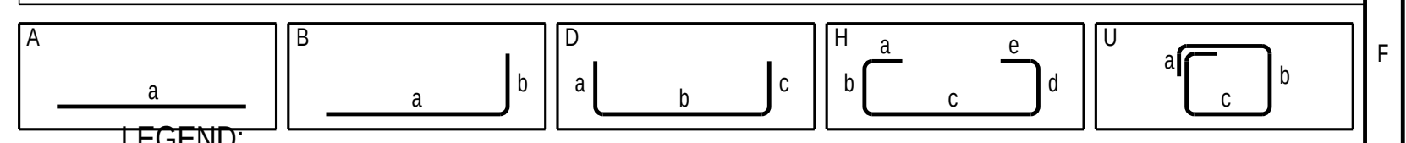
REINFORCING BAR TOTAL WEIGHT [kg] 108.8

REINFORCING BAR LIST — CAST UNIT CODE: P185 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1851	7	B500B	8	1390		3.9	106	320	320	320	320				32	
D	P1852	6	B500B	12	1140		6.1	450	300	450						60	

REINFORCING BAR TOTAL WEIGHT [kg] 9.9

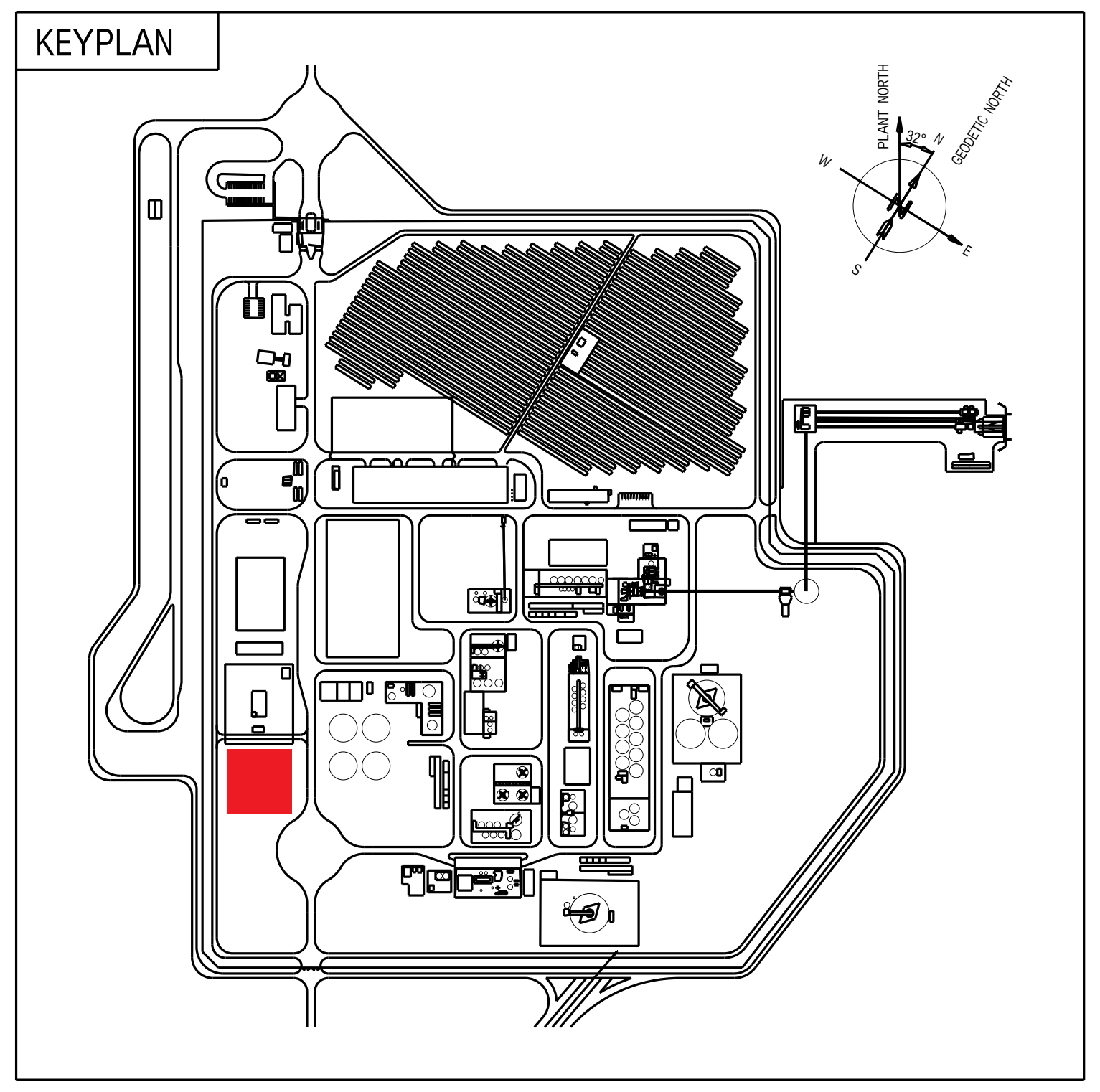
REINFORCEMENT BENDING SHAPES:


A
a
B
a
b
D
a
b
c
H
a
e
b
c
d
U
a
c
b

LEGEND:
R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) UNLESS NOTED OTHERWISE & C15 (15MPa) FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
a) FOUNDATION: 75 mm (BOTTOM & SIDE)
b) FOUNDATION: 50 mm (TOP)
c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DIP GALVANIZED
8. T.O.C. & TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10MM OF DESIGN LEVEL
9. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 375 KN/SQ.M.
10. REBARS CAN BE MOVED LOCALLY BY +/- 25mm TO AVOID INTERFERENCE WITH EMBEDDED PARTS


KEYPLAN
PLANT NORTH
GEODETIC NORTH
32°
N
W

DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-CV-DED-1027	SLAB SECTOR GL G-J/5-10 DETAIL- AND REINFORCEMENT DRAWING		
MD-101-SMEM-EG-ST-DST-0001	CONCRETE STUCTURAL STANDARDS	MD-216-49W1-EG-CV-DED-1026	SLAB SECTOR GL G-J/10-12 DETAIL- AND REINFORCEMENT DRAWING
MD-216-49W1-EG-CV-GAD-1003	1-10/A-E GENERAL ARRANGEMENT DRAWING	MD-216-1000-EG-ST-CRT-1001	DESIGN CRITERIA
MD-216-49W1-EG-CV-GAD-1001	1-8/E-J GENERAL ARRANGEMENT DRAWING	MD-216-49W1-EG-CV-GAD-1002	7-12/D-J GENERAL ARRANGEMENT DRAWING

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	29.01.2024	AS-BUILT	JonRen	MauNot	NanWu	LuiPiz	
C01	01.03.2022	ISSUED FOR CONSTRUCTION	JonRen	MauNot	NanWu	LuiPiz	
A01	18.02.2022	ISSUED FOR APPROVAL	JonRen	MauNot	NanWu	LuiPiz	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER
REVISION 2

OUTOTEC DRAWING NUMBER 900707_GAU01_AG11_CTB01_49037
DOCUMENT ID OU100746294
REVISION 2

DRAWN	JonRen	18.02.2022
CHECKED	MauNot	18.02.2022
ENGINEER	NanWu	18.02.2022
APPROVED	LuiPiz	18.02.2022
EMPLOYER		
SCALE	1:15	A0

MANSOURAH - MASSARAH GOLD PROJECT
49W1 RAW WATER TREATMENT PLANT
SEC. PLATFORMS, STAIR AND PEDESTAL FOUNDATION 2 DETAIL- AND REINFORCEMENT DRAWING

DRAWING NUMBER MD-216-49W1-EG-CV-DED-1037
REV D01